

Goal: Informative Prior

Binary data 1 = defective 0 = not defective

Historically $\mu = 0.05$ $\sigma^2 = 0.0025$

Prior: $\Theta \sim \text{Beta}(a, b)$

$$\mu = \frac{a}{a+b}, \quad \sigma^2 = \frac{ab}{(a+b)^2(a+b+1)}$$

Solution:

$$\frac{a}{a+b} = 0.05$$

$$a = 0.05(a+b)$$

$$0.95a = 0.05b$$

$$\boxed{19a = b}$$

$$0.0025 = \frac{ab}{(a+b)^2(a+b+1)} = \frac{19a^2}{(20a)^2(20a+1)} = \frac{19a^2}{400a^2(20a+1)}$$

$$\frac{0.0025}{1} = \frac{19a^2}{400a^2(20a+1)}$$

$$a^2(20a+1) = 19a^2 \quad a \neq 0$$

$$20a+1 = 19$$

$$20a = 18$$

$$a = \frac{18}{20} = 0.9, \quad b = \frac{19 \cdot 18}{20} = 17.1$$

Prior:

$$\boxed{\Theta \sim \text{Beta}(0.9, 17.1)}$$

Binomial Distribution

$$p(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y} \rightarrow \text{Beta distribution}$$

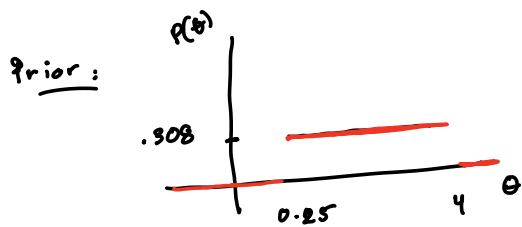
Lesson 1 of module 2: prior $p(\theta) = 1$ for $\theta \in (0,1)$
Beta(a,b)

Lesson 2 of module 2: informative prior

If our prior is a Beta distribution, what is the posterior?

$$P(\theta|y) \propto \underbrace{\theta^y (1-\theta)^{n-y}}_{\text{likelihood}} \underbrace{\theta^{a-1} (1-\theta)^{b-1}}_{\text{prior}}$$
$$= \theta^{y+a-1} (1-\theta)^{n-y+b-1}$$

$\theta|y \sim \text{Beta}(y+a, n-y+b)$



$$p(\theta) = \begin{cases} .308 & \text{for } 0.25 \leq \theta \leq 4 \\ 0 & \text{o.w.} \end{cases}$$

posterior:

$$p(\theta|y) \propto p(\theta) p(y|\theta)$$

$$= \begin{cases} .308 \exp\left(-\frac{(y-\mu)^2}{\sigma^2}\right) & 0.25 \leq \theta \leq 4 \\ 0 & \text{o.w.} \end{cases}$$

prior $\rightarrow$ uniform

posteriors $\rightarrow$??

likelihood $\rightarrow$ normal distribution

So, this is not conjugate